A Survey on Intelligent Transportation Systems for Automated Highway Toll Assortment and Enhanced Security

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***Abstract*— National highways face growing security challenges, including terrorism, illegal transport of hazardous materials, and vehicle theft, necessitating advanced solutions for road safety. This paper presents a three-stage toll gate security system powered by a Raspberry Pi-based platform to automate and enhance security at toll stations. In Stage 1, back-scattered X-ray scanning detects hidden threats within vehicles, with immediate alerts enabled through real-time data processing. Stage 2 employs capacitive stretch sensors to detect vehicle overloading and potential structural modifications indicative of unauthorized transport activities. Stage 3 integrates RFID technology for automated toll collection and GSM communication to notify authorities of toll payments, security threats, and potential vehicle theft, using a database to cross-reference registered RFID tags and identify stolen vehicles. Additional features include GPS-based toll calculations based on distance and LIDAR-based speed detection to enforce speed limits and adjust tolls dynamically. This cost-effective system leverages Raspberry Pi for efficient data processing, providing real-time monitoring, automated alerts, and theft detection, thereby improving security and operational efficiency on national highways.**

Keywords*— *X-Radiation Energy, sensors, raspberry pi, GSM, GPS.

I. INTRODUCTION

In the recent years, the security challenges in national highways have been of grave concern mainly because of the escalating threat of terrorism, illegal transport of hazardous substances, theft of vehicles, and other unauthorized activities. Highways form a lifeline for transportation not only within but also across nations; Hence the need for enhanced security measures is higher than ever before. Traditional security systems have proven incapable of handling the complexities of today's threats and require more sophisticated, automated systems that ensure higher protection and efficiency in operations [1][3].

The proposed three-stage toll gate security system guarantees automated threat detection and decision-making. In Stage 1, a back-scattered X-ray scanner identifies hidden weapons or explosives. If a threat is detected, a GSM alert is sent to the authorities, the alarm is activated, and the exit gate is locked. Otherwise, the vehicle moves to the next stage [1][8]. During Stage 2, capacitive stretch sensors verify for overloading and structural changes. If the vehicle is over the allowed weight limit, an alert is issued to the road transport authority, and the vehicle is routed for checking. If within limits, proceeds to the last stage. During Stage 3, RFID-based toll collection confirms vehicle registration, deducts fares, and identifies stolen vehicles [5][7][10]. GPS-based tolling varies charges depending on distance traveled, while LIDAR tracks speed and levies fines for infringement. Cars completing all three stages depart without interruptions, while intercepted cars are retained for additional security screening, thus providing highway security improved and automated surveillance.

The general architecture of the three-stage toll gate security system is presented in Fig. 1.1, showing the integration of X-ray scanning for threat identification, weight sensors for load confirmation, and RFID technology for automated toll payment. The system further includes GSM communication for real-time notifications and automated gate control at every step to ensure that only cleared vehicles are allowed to move through the toll plaza [1][4][8].

By using real-time monitoring, automated warning, and smart toll processing, this system enhances national highway security, operational efficiency, and quick response to security threats. Using Raspberry Pi for data processing provides a cost-efficient and scalable solution, which makes it ideal for deployment on major highways [1][4].

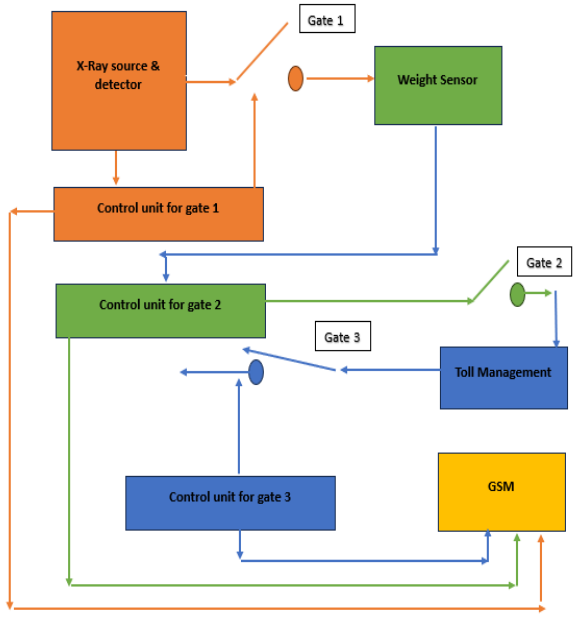

Fig 1.1 Architecture of the three-stage alarming system on road highways

This survey paper is intended to identify and explore innovative solutions. It will be by far a complete study on how they may be implemented in enhancing highway security. This approach through an advanced and highly innovative technology approach, may transform the management and security of toll gates. This becomes a more proactive automated way of addressing growing challenges that highway authorities are facing today.

II. TECHNOLOGIES FOR ENHACED HIGHWAY SECURITY

Improved highway security has called for the development of advanced technological solutions that can automate security measures and efficiently address emerging threats. The proposed system is a combination of various cutting-edge technologies that will bring about a multi-layered proactive approach to highway security, which will provide more protection for drivers and efficient toll operations. These technologies, such as back-scattered X-ray scanning, capacitive stretch sensors, RFID, GPS-based toll calculations, and LIDAR-based speed enforcement, are integrated into a unified platform that allows for real-time monitoring, rapid response to incidents, and seamless user experiences. In this section, we will explore each of these technologies in detail, discussing their functionalities, benefits, integration strategies, and their collective impact on improving highway security.

Back-Scattered X-ray Scanning: Back-scattered X-ray scanning is a technology that is essentially used to find concealed items inside a vehicle like illegal substances, weapons, explosives, or even contrabands that may prove to be hazardous. This low-energy X-rays penetrate the vehicle and bounce off the objects that are inside to show a detailed picture of what can be hidden within without opening the vehicle. It offers a non-intrusive method of scanning the vehicle at the toll gates in real-time as they pass through, integrating back-scattered X-ray scanners. This system can automatically flag suspicious items, ensuring quick identification and investigation of vehicles posing a potential threat. This ensures that there is no risk of manual errors or human oversight but maintains a smooth flow of traffic. The advancing technology of X-ray scanning means that resolution and accuracy in scans will increase and be even more effective at detection without compromising efficiency.

The main advantage of back-scattered X-ray scanning is that it can scan rapidly and thoroughly without having to stop or halt traffic and create bottlenecks; this comes in very handy for busy highways. The major challenge, however, will be to ensure the system accurately distinguishes between threats and harmless items because false positives may be a cause for piling up unnecessary delays. This can be done by continuous improvement in algorithms for image recognition as well as improved calibration of the system [1].

Capacitive Stretch Sensors: Capacitive stretch sensors detect physical deformations or strain caused by objects passing over them, such as vehicles with abnormal loads or structural alterations. These sensors can be embedded in the roadway or installed at key checkpoints, such as toll booths, to monitor vehicle conditions in real-time. Overloaded vehicles pose a significant risk to both road safety and infrastructure, as they can cause accidents, increased wear on road surfaces, and damage to bridges and overpasses. Capacitive stretch sensors can automatically alert authorities to potentially dangerous vehicles by detecting weight imbalances and irregular structural changes. These sensors are highly sensitive and help detect minute variations in weight between the vehicles. This is crucial for the safety of the highway system in maintaining proper ordering [1].

In terms of integration, capacitive stretch sensors can be cross-referenced in real time with other sensor inputs such as RFID and GPS for identification of the vehicle that would necessitate further inspection or enforcement action. Vehicle loads may now be remotely and instantaneously assessed, lessening the load on human inspection teams that can be prioritized for higher-priority cases. The main drawback of this technology is interference with environmental factors such as road surface conditions or extreme weather, which may affect sensor accuracy. Calibration and maintenance must be done to ensure that capacitive stretch sensors remain reliable and accurate in the long term [6].

RFID Technology for Automatic Toll Collection: The efficient and automated collection of toll fees requires RFID technology as a critical enabler. This is possible through RFID tags attached to vehicles, where the system can automatically detect and identify vehicles as they pass through toll booths without requiring drivers to stop and pay manually Automation reduces congestion at toll stations, speeds up the flow of traffic, and improves the overall travel experience for highway users. RFID systems are also very effective in managing large volumes of vehicles because they can read tags quickly and accurately even in heavy traffic conditions [

Beyond toll collection, RFID technology can also support security by tracing the movement of vehicles in real-time. Such information can help identify suspicious or vehicles of interest, such as those with stolen plates or violators of traffic laws. Vehicle movement tracking will enable enforcement agencies to respond quickly to security threats, thereby improving the system's ability to detect and respond to illegal activities on the highway. A challenge with RFID technology is data privacy and protection against misuse, such as unauthorized tracking of individual vehicles. This can be achieved by implementing strong encryption and privacy protocols to protect collected data.

GPS-Based Toll Calculation: The GPS technology enables the dynamic calculation of tolls depending on the distance a vehicle covers on the highway. This contrasts with the traditional system, where tolls are collected as a flat rate. GPS-

based systems offer a more accurate pricing model, which incorporates factors such as route choice, traffic conditions, and time of travel. This makes for a more flexible and fair tolling system that can adapt to changing loads or security issues. GPS also provides real-time location tracking, which can be helpful in tracking vehicle movement and identifying possible security risks.

Enhanced situational awareness will be a benefit to highway security in the use of GPS-based toll systems because authorities will be able to track the locations and speeds of vehicles over long distances. When a vehicle takes an inappropriate detour or is identified as speeding through a high-risk area, law enforcement will be notified in real time, thus enabling an immediate response. The GPS data also be used for the optimization of traffic flow through the dynamic variation of toll prices based on congestion at peak hours. However, it is reliant on GPS signals and may encounter issues in tunnels or densely built environments where the satellite coverage is weak. To reduce such problems, supplementary technologies such as road sensors or cellular-based positioning can be employed to continue the monitoring.

LIDAR-Based Speed Limit Enforcement: LIDAR technology employs laser pulses to measure the distance in extremely high resolution and, thus, is well suited for real-time vehicle speed detection. In contrast to radar-based systems, LIDAR is more accurate and can monitor the speed of single vehicles even in heavy traffic. The incorporation of LIDAR technology in highway security systems enables dynamic enforcement of speed limits according to real-time traffic conditions or environmental factors, like weather, road surface, and congestion. This adaptability is critical to ensuring that vehicles travel at safe speeds and reducing the number of accidents due to speeding.

LIDAR can be installed at different points along roads to measure traffic speeds all the time, automatically issuing tickets or warnings for speeding. The major benefit of LIDAR is its precision, but the difficulty is its restricted range and the possibility of interference in poor weather, e.g., heavy rain or fog. To mitigate these problems, LIDAR devices need to be precisely calibrated, and multiple sensors can be installed to survey wider areas and maintain continuous surveillance. In addition, combining LIDAR with other technologies, including RFID and GPS, can facilitate more effective enforcement by linking vehicle speeds to other data points, including route and behavior patterns.

System Integration and Data Flow: The true strength of this security system is in how it brings all these disparate technologies under one unified platform. By connecting back-scattered X-ray scanners, capacitive stretch sensors, RFID systems, GPS-based toll computation, and LIDAR to a centralized control center, the system can process information from various sources in real time. The centralized flow of data makes it possible for the system to make automated decisions, such as notifying authorities of potential security risks, enforcing traffic regulations, or dynamically varying toll rates based on traffic flows. The Raspberry Pi platform ensures that the system is affordable and scalable, making it easy to deploy across an extensive network of toll booths and highway checkpoints.

Although this integrated system has significant advantages in security, efficiency, and real-time observation, it is also fraught with difficulties in managing data, ensuring system reliability, and scalability. For example, data processing and analysis of a large amount of data from numerous sensors might put a heavy burden on computing resources if not well optimized. Moreover, guaranteeing data accuracy and synchronization between the different technologies is crucial for preserving the integrity of the security system. Still, the potential rewards of an integrated system that optimizes both security and traffic movement make this strategy an appealing option for contemporary highway networks.

III. LITERATURE SURVEY

Terrorism, hijacking of cars, and transportation of harmful materials are some of the critical issues facing national highways in terms of security. For the above purposes, much research has offered solutions in the technological dimension, primarily incorporating IoT devices, sensors, and automation technologies, which significantly enhance threat detection capabilities and allow quicker response time for law enforcement agencies and enhancement of toll management with more efficiency.

From highway security literature, it is quite evident that numerous advanced technologies are involved in the design, such as back-scattered X-ray scanning for threat detection, capacitive stretch sensors to monitor vehicles, RFID for automated toll collection, GPS-based systems for fine-tuned pricing at tolls, and LIDAR for enforcing speed. As integrated into platforms, these technological advancements offer excellent solutions to problems in security and operational challenges highway authorities face.

A comprehensive summary of key research papers in this area is presented in the following table. It shows technologies used, advantages, and any limitations or challenges associated with each approach, providing insight into effectiveness in highway security concerns.

TABLE I. Summary of Technologies for Highway Security and Operational Efficiency

PAPER	TECHNOLOGY USED	ADVANTAGES	DISADVANTAGES
Three stage Toll Gate Alarming mechanism	Capacitive Stretch sensors, RFID, GSM	Multistage security (weapons, overload detection toll automation), high reliability	Complex architecture, high implementation cost
Advance Toll System with Automatic Vehicle Overweight Detection.	RFID, Load cells, ESP32, IOT	Real time overweight detection, integration with cloud, efficient traffic management	Limited to prototypes, scalability issues for large systems
IOT, RFID and Cloud systems	IOT, RFID, Cloud	Real time toll collection and monitoring with reduced human dependency	Requires large scale infrastructure changes and has data security issues
RFID based Toll Gate Management System Using IOT Monitoring	RFID, Raspberry Pi, ANPR Camera, Cloud, and stepper Motor	Automated distance-based toll collection system with the ability to process transactions quickly and provide fair pricing to the users according to the distance covered.	Hardware precision, Constant internet connectivity
Automated Toll Gate System Using RFID And GSM Technology	RFID, GSM, IR Sensors	Cost effective toll management with real time updates	IR Sensors malfunction in harsh
Design of an Integrated Speed Tracking and Toll Management System for Express Highways	RFID, Raspberry Pi, Ultrasonic Sensors, ANPR Camera, Cloud Storage, and a Buzzer.	Automatic vehicle identification, monitors speed, and manages toll collection efficiently.	Sensor precision, steady power supply, internet connectivity, and good RFID tag visibility.
Automatic Toll Collection Using QR Code	Barcode scanners, RFID	Simple and inexpensive to implement	Inefficient for moving vehicles, lacks advanced tracking
An Automated Electronic Toll Collection system using Zigbee	RFID, Zigbee transceivers	Reduces manual errors and enables greater communication range	Interference sensitivity and limited Zigbee range
Location Based Efficient Road Toll Collection System using RFID and NPRS Techniques	RFID, ANPR Camera (NPRS), GPS, Raspberry Pi, Cloud, and Stepper Motor.	Reduced manual work, Quicker transactions, Reasonable pricing according to distance covered, Improved data management and access through cloud integration	Very much hardware accuracy and internet-dependent, Needs frequent maintenance, can be problematic if cars do not have proper RFID tags or number plates are not legible
A Cost-Effective Security Technology Integrated with RFID Based Automated Toll Collection System	RFID Reader, RFID Tag, Arduino Mega, Force Sensitive Resistor (FSR) LCD Display, GSM Module Servo Motor	Improved security, Automatic toll collection, Real-time communication, Time and cost effective	Limited RFID range, Prone to junk messages, Manual recording of data, Dependence on GSM network
Theft Detection and Location Tracking in Electronic Toll Collection System Using RFID and AES Algorithm	RFID, AES Algorithm, Arduino, Wi-Fi, GPS, IOT	Decreases traffic congestion, reduces fuel consumption, online transactions, Identifies and blocks stolen cars, Secures data via encryption	High initial expense, Internet and power supply dependent, RFID signal interference or duplication feasible

IV. OPEN CHALLENGES AND FUTURE DIRECTIONS

While the proposed three-stage toll gate security system enhances highway safety, there are many challenges to overcome. One such challenge is scalability, with cost-effective deployment being necessary when applied to many highways and ensuring a smooth integration into current infrastructure. Anomaly detection using AI can be considered for enhancing X-ray scanning and RFID-based vehicle verification to eliminate false positives and enhance threat identification accuracy. Blockchain-based data protection can enhance data integrity, providing tamper-proof storage of vehicle credentials, toll transactions, and security logs. Future development in X-ray imaging technology will further enhance the detection of hidden weapons and dangerous materials, enhancing security screening. Integration with national crime databases will also facilitate real-time stolen vehicle identification using RFID and GSM communication. Enhancements in LIDAR-based speed enforcement and GPS-based tolling will provide real-time dynamic adjustments of toll rates based on real-time traffic situations, allowing the system to become more intelligent and efficient. The resolution of these issues will pave the way to create a stronger, more scalable, and wiser national highway toll security system.

V. CONCLUSION

From the above research, the three-stage toll gate security system introduced here is an innovative and effective solution for improving national highway security and operational efficiency. The system incorporates back-scattered X-ray scanning, capacitive stretch sensors, RFID-based tolling, GPS-based pricing, and LIDAR speed enforcement, all based on a cost-effective Raspberry Pi platform. The integration of these technologies provides real-time monitoring, automated threat detection, and proactive security enforcement.

Although the system brings major improvements in highway safety, some issues remain, such as scalability, sensor integration, and optimization of real-time data processing. These elements are important to address for smooth deployment. The system's capability to identify concealed threats, deter vehicle theft, and control traffic indicates its potential to transform highway security. Nevertheless, additional research and development are needed to improve reliability, flexibility, and large-scale implementation.

This survey emphasizes the necessity of smart and technology-enabled security solutions to ensure safe, effective, and automated surveillance of national highways. The system suggested here not only solves present security issues but also opens avenues for a more advanced and stronger highway management system in the future.

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